

Changgeon Kang

Ph.D. Student at KAIST

 linkedin kangha02@kaist.ac.kr

Research Interests

Cyber-Physical Systems (CPS) Security, Sensing systems

My research focuses on discovering novel physical-layer vulnerabilities in IoT and sensing systems. I am particularly interested in exploiting unintentional wireless signal leakage, such as backscatter, to remotely extract privacy-sensitive data.

Education

KAIST

Ph.D. in School of Computing (Advisor: Jun Han)

Daejeon, Korea Sep 2026 - present

KAIST

M.S. in School of Computing (Advisor: Jun Han)

Daejeon, Korea Sep 2024 - Oct 2026

Yonsei University

B.S. in Electrical and Electronic Engineering

Seoul, Korea Mar 2018 - Feb 2024

Publication

Full Paper

- “ScatterPIN: PIN Inference Attack on Touch Keypad Door Locks via RF Backscatter Side Channel”

Changgeon Kang, Dongjin Seo, Jonghyuk Yun, Seungbeom Park, Junyong Kim, and Jun Han

IEEE Symposium on Security and Privacy (IEEE S&P) 2027. Montreal, Canada. *(To appear)*

Poster

- “Poster: Exploiting Keystroke Dynamics via mmWave Radar for Application Profiling”

Changgeon Kang, Dongjin Seo, Sihun Yang, and Jun Han

ACM International Conference on Mobile Systems, Applications, and Services (ACM MobiSys) 2024. Tokyo, Japan.

Patent

Domestic Patent Application

- Security Vulnerability Analysis of Door Locks Using Backscatter from Circuit Signals [10-2025-0140616]

Jun Han, Changgeon Kang, Dongjin Seo, Yeonjun Lee

Research Experience

Graduate Researcher, KAIST

Sensing Intelligence and Cyber-Physical Security (CyPhy) Lab

Daejeon, Korea Sep 2024 - present

Research Assistant, KAIST

Sensing Intelligence and Cyber-Physical Security (CyPhy) Lab

Daejeon, Korea Feb 2024 - Oct 2024

Research Assistant, Yonsei University

Cyber-Physical Systems and Security Lab

Seoul, Korea Dec 2022 - Jan 2024

Teaching Experience

Project Mentor

Mobile Computing and Applications (CS442), KAIST

Spring 2024

Internet-of-Things Security (IS593), KAIST

Fall 2024

Special Topics – Introduction to Sensing Security (CS492), KAIST

Spring 2025

Teaching Assistant

Special Topics – Introduction to Sensing Security (CS492), KAIST

Spring 2026

Honor and Awards

- National Science and Technology Scholarship (2018-2023)
- Semester Honor, Yonsei University (Fall 2019, Spring 2020)
- Semester High Honor, Yonsei University (Fall 2020)

Professional Service

External Reviewer

- **Mobile Computing Conferences** (ACM MobiCom'25/26, ACM MobiSys'24/25/26, ACM SenSys'24/25/26/27, ACM/IEEE IPSN'24)
- **Security Conferences** (USENIX Security'26, ACM CCS'24, IEEE S&P'24/25/27, ACM WiSec'26)
- **HCI Conferences** (ACM UIST'25)
- **Workshops** (ACM HotMobile'24/25/26)

Skills

- **Programming Language:** Python, C, C++
- **Tools and Framework:** GNU Radio, Matlab, mmWave Studio